

Vincent Tang

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EDUCATION

Northwestern University

Evanston, IL

Bachelor of Science in Computer Science, GPA: 3.66/4.0

Expected June 2029

- Relevant Coursework: Software Design (C/C++), Data Structures & Algorithms, Machine Learning, Software Quality Engineering, Large Language Models, Software Design Principles, Knight Lab Studio

EXPERIENCE

Software Engineering Intern

May 2026 – September 2026

Raiders Marketplace

Remote

- Refined the UI/UX of a SwiftUI/MVVM iOS marketplace app used by 100+ users and cut content load time ~70% by implementing image and category-metadata caching layers
- Built a real-time sync pipeline (Swift, PostgreSQL, Supabase) with database triggers, row-level security (RLS), and live channels, and fixed event loss from RLS and authentication race conditions
- Scaled the app from 1 to 3 seller types by building reusable SwiftUI components, polymorphic JSONB scheduling, and MapKit location selection, validated on a dedicated staging database

Hackathon Program Intern

Dec. 2025 – March 2026

Grassroot Academy

Evanston, IL

- Designed and launched a 3-week remote hackathon from scratch, guiding 5 teams through the full software development lifecycle from ideation to final presentation
- Hosted 3 technical workshops with guest speakers, coordinated team milestones, and mentored beginner developers, helping teams ship original projects on time

Computer Consulting Aide

Oct. 2025 – Present

NU Student Affairs IT

Evanston, IL

- Supported 25+ Student Affairs departments by diagnosing and resolving hardware, software, and network issues across desktops, laptops, printers, and office phones, minimizing downtime
- Triageed and resolved an average of 6–8 IT support tickets per week, identifying root causes and implementing efficient fixes to restore system functionality

PROJECTS

Genshin Combat Optimization Engine | *Next.js, React, TypeScript, Vitest, Playwright*

September 2026

- Built a data-driven modeling pipeline unifying 130+ characters, 245+ weapons, 60+ artifact sets, and 120+ artifact effects, enabling team and build simulations with no hard-coded loadouts
- Engineered a deterministic optimizer that evaluates 300+ action sequences in under 900ms, surfacing one that deals ~16% more damage than the baseline (160K+ vs. 138K)
- Built a simulation engine modeling action timing, energy, elemental reactions, buffs, and damage attribution, validated by 800+ passing tests and an end-to-end Playwright simulation

CareerPilot | *Python, discord.py, Supabase, FastAPI, Google GenAI, Docker, BeautifulSoup4, GitHub API*

July 2026

- Engineered a job-ingestion pipeline that pulls 10K+ postings (6.5K+ internships, 4.2K+ new-grad roles) from 5 sources every 15 minutes, then normalizes, deduplicates, and stores enriched records in Supabase
- Built a two-stage resume-to-job matcher scoring 5 job-fit dimensions in Python with deterministic must-have constraints, validated on all 5 regression scenarios across 15 live model runs
- Containerized and deployed a FastAPI/Tectonic resume compiler that generates one-page PDFs, replacing a ~5GB TeX Live dependency with a ~50MB binary while enforcing auth, request limits, and compile timeouts

Exploding Kittens | *Java, JavaFX, JUnit, Gradle, GitHub Actions, TDD, MVC*

June 2026

- Built a Java/JavaFX multiplayer card game (2–5 players, 16 card types) with a 4-person team using TDD, boundary-value analysis, Checkstyle, SpotBugs, and PR reviews
- Reached a ~95% PIT mutation score across 250+ JUnit/Cucumber tests, with automated Gradle CI enforcing lint, static analysis, and coverage on every PR
- Owned the entire UI/UX and frontend (5 JavaFX screens), creating and refining the interface and wiring its MVC layer to a shared GameEngine module so UI and engine could be developed and tested independently

TECHNICAL SKILLS

Languages: Python, Java, TypeScript, Swift, C/C++, Objective-C++, JavaScript, SQL, HTML/CSS

AI-Assisted Development: Codex, Claude, Cursor

Frameworks & Libraries: Next.js, React, React Router, Node.js, Express.js, FastAPI, discord.py, BeautifulSoup4, Google GenAI, SwiftUI, Tailwind CSS, PyTorch, scikit-learn, NumPy, Pandas, Matplotlib, Seaborn

Developer Tools & Platforms: Git, GitHub, GitHub API, GitHub Actions, Gradle, JUnit, Cucumber, Vitest, Playwright, Checkstyle, SpotBugs, PostgreSQL, Supabase, Docker, Vercel